

# Zhuangze Hou

Ph.D. Candidate in Creative Media | City University of Hong Kong | Hong Kong, China

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## EDUCATION

- Ph.D. Candidate in Creative Media · City University of Hong Kong** Jan 2025–Expected Jan 2029  
*School of Creative Media · Supervisor: Prof. Can Liu · Research area: Human–Computer Interaction.*
- Master’s Degree in Software Engineering · Dalian University of Technology** 2021–2024  
*Supervisor: Prof. Qiufen Xia · Research on VR interaction, visual guidance, and intelligent systems.*
- Bachelor’s Degree in Software Engineering · Dalian University of Technology** 2017–2021  
*Majored in Software Engineering.*

## RESEARCH INTEREST

- Multimodal Large Models: multimodal intent understanding, spatial reasoning, and context-aware interaction.
- Multimodal and Embodied Interaction: speech, gesture, gaze, body movement, object manipulation, and interaction history.
- Generative AI for Embodied Creation: progressive human–AI co-creation, editable and reversible generation, and user embodied input and control.
- VR/MR and Spatial Computing: immersive content creation, 3D scene manipulation, spatial interaction, and MR physical making.

## PUBLICATIONS

- Zhuangze Hou, Jingze Tian, Nianlong Li, Farong Ren, and Can Liu. “EchoLadder: Progressive AI-Assisted Design of Immersive VR Scenes.” ACM UIST 2025. First author.**  
*Contribution: Contributed to the research concept; built the vision–language-model-to-Unity pipeline and interaction system; led the user study, analysis, and writing.*
- Yangyang He\*, Zhuangze Hou\*, Yonglin Chen, and Can Liu. “Reference-Based Manipulation: A Framework and Pipeline for Multimodal Spatial Reasoning.” ACM UIST 2026. Co-first author.**  
*Contribution: Co-developed the Source–Anchor–Frame representation and multimodal reasoning pipeline; contributed to implementation, experiments, analysis, and writing. \* Equal contribution.*
- Zhuangze Hou, Qiufen Xia, Zichuan Xu, and Yidi Tang. “Arradar: Arrow and Radar Driven HMD-Based Visual Guidance in Limited Field-of-View.” IEEE ISMAR-Adjunct 2023 (Poster). First author.**  
*Contribution: Conceived and led the project; designed and implemented the Unity guidance prototype; led the experiment design, data analysis, and writing.*

## PATENT

- Qiufen Xia, Zhuangze Hou, and Zichuan Xu. “Method and System for Adjusting User Field of View (FoV) in a Three-Dimensional Virtual Reality Scene.” Granted invention patent · CN114779940B (2024) · US11880952B2 (2024). Second inventor.**  
*Contribution: Contributed to the VR field-of-view adaptation method that uses projected scene flow to reduce discomfort while preserving immersion.*

## ENGINEERING & PROFESSIONAL EXPERIENCE

- Technical Staff · Hong Kong Productivity Council** Nov 2024–Sep 2025
- Supported technical projects through data collection, documentation, report preparation, and project coordination; explored AI applications in energy and low-altitude-economy scenarios.
- Computer Vision & Edge AI · AI Management Platform Project** 2021–2024  
*Dalian University of Technology*
- Developed and trained visual detection models, including safety and industrial-scene tasks, and contributed to model quantization and edge deployment.

## **SELECTED AWARDS**

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- Outstanding Master's Thesis, Dalian University of Technology · University-level award, top 10% · Jun 2024
- Third Prize, 9th MathorCup University Mathematical Modeling Challenge · Jun 2019

## **SERVICE**

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- IMWUT 2026 Reviewer